

Algebraic Proof Scramble 2.4 · Team Challenge

Teams of 4. Nothing is written until Round 3. Printing: each strip page is ONE pile — guillotine straight across, drop the page in the envelope.

Roles *rotate each round*

SORTER lays out the strips. NAMER matches reasons. CHECKER runs Rule 4 on every answer. CAPTAIN calls it done — and only the Captain may raise a hand.

Scoring

Round 1 — 6 pts Round 2 — 4 pts Round 3 — 5 pts Round 4 — 10 pts
First team correct in a round: +2. A wrong call: -1 and you go back to work. Captain raises a hand ONCE per round.

Round 1 — Sort and Build *envelope 1 · three proofs, all mixed together*

Three proofs are in one pile. Sort the statements first — each proof uses a different letter. The reasons do not sort that way; you have to match them by what they do.

SELF-CHECK *TWO reason strips belong to no line at all. If nothing is left over, one of your reasons is in the wrong place. Also: your three answers add to 9.*

Round 2 — Shortest Legal Proof *no strips · whiteboards*

Every team gets the same equation: $5(x + 3) = 40$. Fewest LINES wins. Every line still needs a legal reason.

SELF-CHECK *Three lines is possible. If you distributed first you used four — look again at what you could do to BOTH sides.*

Round 3 — Missing Link *envelope 3 · one strip has been removed*

This proof is missing one statement. Build what you can, then WRITE the missing line and name its reason.

SELF-CHECK *The strips leave exactly one hole. If you have two holes, something is out of order.*

Round 4 — The Closer *envelope 4 · nine lines · worth double*

An algebraic proof wearing geometry clothes. Two of these reasons are NOT properties of equality.

SELF-CHECK *The last line must be $AB = 42$. Every strip is used — no ringers in this envelope.*

Envelope 1 — Round 1 *cut every strip apart · one set per team*

STATEMENTS (three proofs, mixed)

$$-6m = 18$$

$$m = -3$$

$$4y + 4 = 24$$

$$4y = 20$$

$$x = 7$$

$$-3(2m - 5) = 33$$

$$7y + 4 = 3y + 24$$

$$5x + 8 = 43$$

$$5x = 35$$

$$-6m + 15 = 33$$

$$y = 5$$

REASONS (two of these belong to nothing)

Reflexive Property of Equality

Subtraction Property of Equality

Subtraction Property of Equality

Division Property of Equality

Given

Distributive Property

Subtraction Property of Equality

Multiplication Property of Equality

Given

Division Property of Equality

Subtraction Property of Equality

Given

Division Property of Equality

Envelope 3 — Round 3 *one statement strip is deliberately missing*

Prove: $2(3k + 1) = 4k + 14 \rightarrow k = 6$

STATEMENTS

$$2k = 12$$

$$k = 6$$

$$6k + 2 = 4k + 14$$

$$2(3k + 1) = 4k + 14$$

REASONS

Subtraction Property of Equality
Subtraction Property of Equality
Division Property of Equality
Given
Distributive Property

Envelope 4 — Round 4 · The Closer *nine lines · every strip is used*GIVEN: M is the midpoint of AB, $AM = 4x - 3$, $MB = 2x + 9$ PROVE: $AB = 42$ **STATEMENTS**

$$4x - 3 = 2x + 9$$

$$2x - 3 = 9$$

$$AM = MB$$

$$AB = AM + MB$$

$$x = 6$$

$$AB = 42$$

$$AM = 21 \text{ and } MB = 21$$

M is the midpoint of AB

$$2x = 12$$

REASONS

Definition of midpoint

Substitution Property of Equality

Subtraction Property of Equality

Substitution Property of Equality

Segment Addition Postulate

Addition Property of Equality

Substitution Property of Equality

Given

Division Property of Equality

Team Record Sheet *Captain writes · one per team*

Team: _____ Members: _____

ROUND 1 $x =$ _____ $y =$ _____ $m =$ _____ the two ringers: _____

ROUND 2 Write your shortest proof:

#	Statement	Reason
1		
2		
3		
4		

ROUND 3 The missing line: _____ Reason: _____

ROUND 4 Write the finished proof:

#	Statement	Reason
1		
2		
3		
4		
5		
6		
7		
8		
9		